

HONGBO WEN

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I am a Ph.D. candidate in Computer Science at UC Santa Barbara, advised by Yu Feng. My research develops program analysis and synthesis techniques to secure complex software systems, with applications in LLM agent security, cryptography & zero-knowledge proofs, and AI-augmented vulnerability discovery. I have discovered zero-day vulnerabilities across LLM agent platforms, DeFi protocols, and ZK circuits, and co-founded Riema Labs as Principal Architect, leading a 50-person engineering team shipping security-critical production systems. My work has been funded by Ethereum Foundation and conducted in collaboration with Google Security and a16z crypto.

EDUCATION

University of California, Santa Barbara, CA, USA
Ph.D. candidate in Computer Science (expected 2027)
GPA 4.0/4.0

Sep 2022 - Present

Tsinghua University, Beijing, China
Bachelor of Engineering

Aug 2015 - Jun 2020

SELECTED PUBLICATIONS

1. **Hongbo Wen**, Ying Li, Hanzhi Liu, Chaofan Shou, Yanju Chen, Yuan Tian, Yu Feng. SEMIA: Auditing Agent Skills via Constraint-Guided Representation Synthesis. arXiv 2026.
2. Hanzhi Liu, Chaofan Shou, Xiaonan Liu, **Hongbo Wen**, Yanju Chen, Ryan Jingyang Fang, Yu Feng. Synthesizing Multi-Agent Harnesses for Vulnerability Discovery. arXiv 2026.
3. Hanzhi Liu, Chaofan Shou, **Hongbo Wen**, Yanju Chen, Ryan Jingyang Fang, Yu Feng. Your Agent Is Mine: Measuring Malicious Intermediary Attacks on the LLM Supply Chain. arXiv 2026.
4. Junrui Liu, Jiaxin Song, Yanning Chen, Hanzhi Liu, **Hongbo Wen**, Luke Pearson, Yanju Chen, Yu Feng. Tabby: A Synthesis-Aided Compiler for High-Performance Zero-Knowledge Proof Circuits. OOPSLA 2025.
5. **Hongbo Wen**, Hanzhi Liu, Yanju Chen, Jingyu Ke, Dahlia Malkhi, Yu Feng. Thunderbolt: Fast Asynchronous Off-Chain Bitcoin Transfers. IACR ePrint 2025.
6. **Hongbo Wen**, Hanzhi Liu, Jiaxin Song, Yanju Chen, Wenbo Guo, Yu Feng. FORAY: Towards Effective Attack Synthesis against Deep Logical Vulnerabilities in DeFi Protocols. ACM CCS 2024.
7. **Hongbo Wen**, Jon Stephens, Yanju Chen, Kostas Ferles, Shankara Pailoor, Kyle Charbonnet, Isil Dillig, Yu Feng. Practical Security Analysis of Zero-Knowledge Proof Circuits. USENIX Security 2024.

INVITED TALKS

Tabby: A Synthesis-Aided Compiler for High-Performance Zero-Knowledge Proof Circuits	Feb 2026
SoCalPLS	USC, Los Angeles, CA, USA
Tabby: Automated Programming of Efficient Zero-Knowledge Proof Circuits	Feb 2025
ETHDenver	Denver, CO, USA
Practical Security Analysis of Zero-Knowledge Proof Circuits	Feb 2025
SoCalPLS	UCSD, San Diego, CA, USA
FORAY: Towards Effective Attack Synthesis against Deep Logical Vulnerabilities in DeFi Protocols	Oct 2024
ACM CCS '24	Salt Lake City, UT, USA
Practical Security Analysis of Zero-Knowledge Proof Circuits	Aug 2024
USENIX Security '24	Philadelphia, PA, USA

SELECTED GRANTS

Ethereum Foundation Academic Grants Round 2024	Jun 2024
Yu Feng, Hanzhi Liu, Hongbo Wen	
Topic: Sentinel: Adaptive Counter-Attack Synthesis for Mitigating Onchain Exploits	
Ethereum Foundation Academic Grants Round 2023	Jun 2023
Yu Feng, Yanju Chen, Hongbo Wen	
Topic: Financial Model-Driven Attack Synthesis for DeFi	

SELECTED ZERO-DAY DISCOVERIES

1. **LLM Agents:** Filed 17 zero-day security advisories on OpenClaw/ClawHub agent skills, covering credential leakage, RCE, and PII exfiltration.

2. **DeFi:** Discovered 10 zero-day vulnerabilities on BNB Chain DeFi protocols using FORAY, with verified exploit traces matching $\geq$ \$21M historical-loss patterns.
3. **ZK Circuits:** Discovered zero-day vulnerabilities in widely-used Circom projects using ZKAP, including 0xPARC, Polygon-ZK, Axiom, iden3, privacy-ethereum.

WORK EXPERIENCE

Riema Labs Inc.

Jan 2024 - Present

Co-founder & Principal Architect.

Highlights:

- Led a 50-person engineering team through an AI-first development transformation, serving 150K+ users.
- Shipped security-critical production systems including cross-chain bridges and ZK infrastructure.

Veridise Inc.

Mar 2022 - Aug 2022

Research & Development Engineer: smart contract and zero-knowledge proof security.

Highlights:

- Discovered 30+ critical vulnerabilities in widely-used ZK projects, and 10+ zero-day DeFi vulnerabilities.

SERVICE

USENIX Security '25 Artifact Evaluation Committee

2025

CS162 Programming Languages, Teaching Assistant, UC Santa Barbara

2023 – 2025

SKILLSET

Languages: Rust, Go, Solidity, C/C++, JavaScript, Python.

AI-Assisted Research: Several projects and publications herein were co-developed with LLM coding agents, directed and verified by me.